CS5275 – The Algorithm Designer’s Toolkit (S2 AY2025/26)

Lecture 12:

Boolean functions – social choice functions

Social choice functions / voting rules

A Boolean function

$$f : \{-1,1\}^n \to \{-1,1\}$$

can be viewed as a **voting rule** for an election with $\underbrace{\text{2 candidates}}_{\{-1,1\}}$ and $\underbrace{n \text{ voters}}_{\text{The input } x \in \{-1,1\}^n \text{ represents their votes.}}$.

also known as social choice function

Common voting rules

Majority: $\mathrm{Maj}_n(x) = \mathrm{sgn}(x_1 + x_2 + \cdots + x_n)$

$$\mathrm{sgn}(s) = \begin{cases} 1 & s > 0 \\ -1 & s < 0 \\ \text{can be 1 or } -1 & s = 0 \end{cases}$$

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also known as **linear threshold function** $a_0, a_1, \dots, a_n \in \mathbb{R}$

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Special cases of weighted majority

The ith dictator function: $\chi_i(x) = x_i$

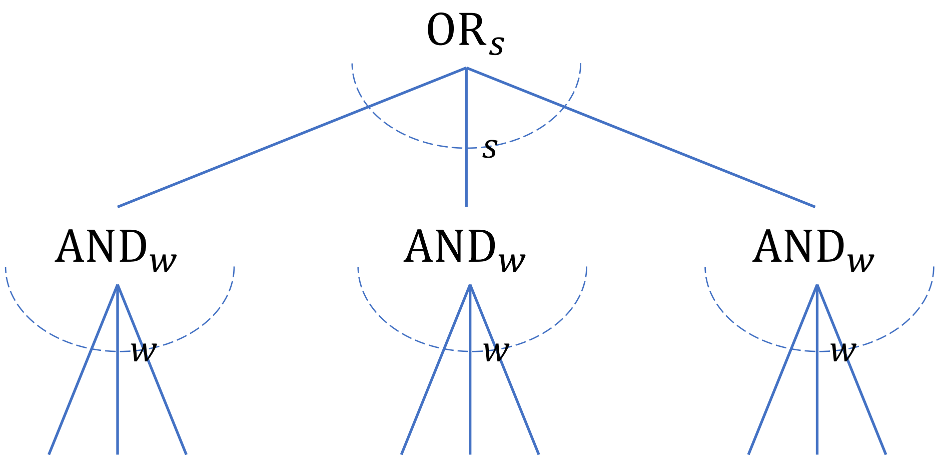
AND: $\mathrm{AND}_n(x) = \begin{cases} -1 & x = (-1, \ldots, -1) \\ 1 & \text{otherwise} \end{cases}$

OR: $\mathrm{OR}_n(x) = \begin{cases} 1 & x = (1, \ldots, 1) \\ -1 & \text{otherwise} \end{cases}$

$-1 = \text{true}$
$1 = \text{false}$

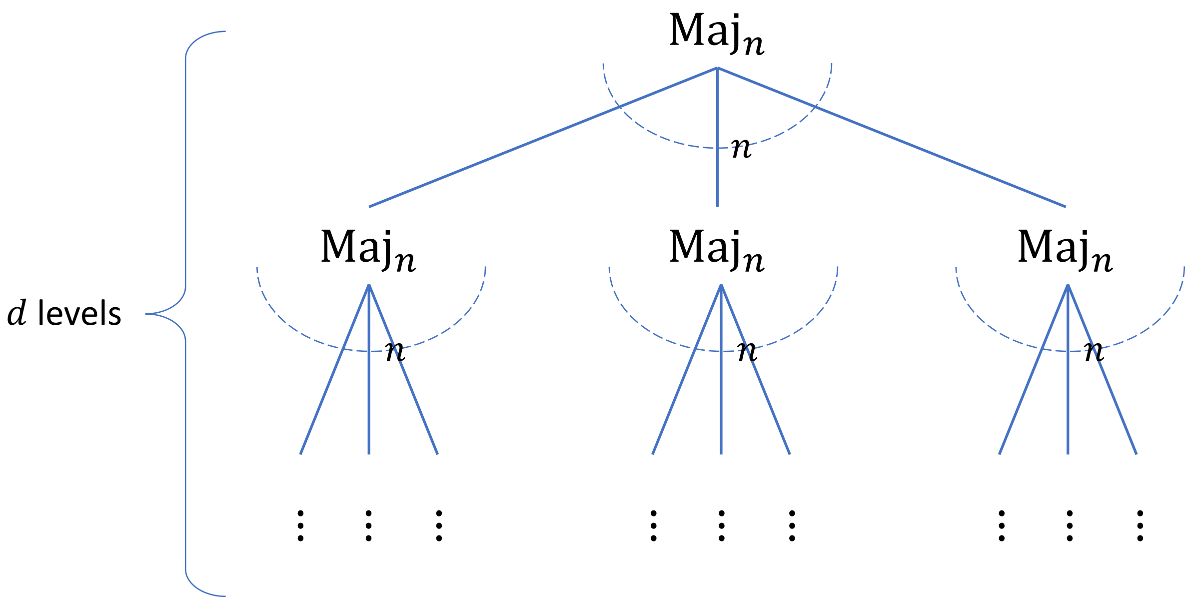
Common voting rules

Tribes: $\mathrm{Tribes}_{w,s} : \{-1,1\}^{sw} \to \{-1,1\}$



Common voting rules

Depth-d majority: $\mathrm{Maj}_n^{\otimes d} : \{-1,1\}^{n^d} \to \{-1,1\}$



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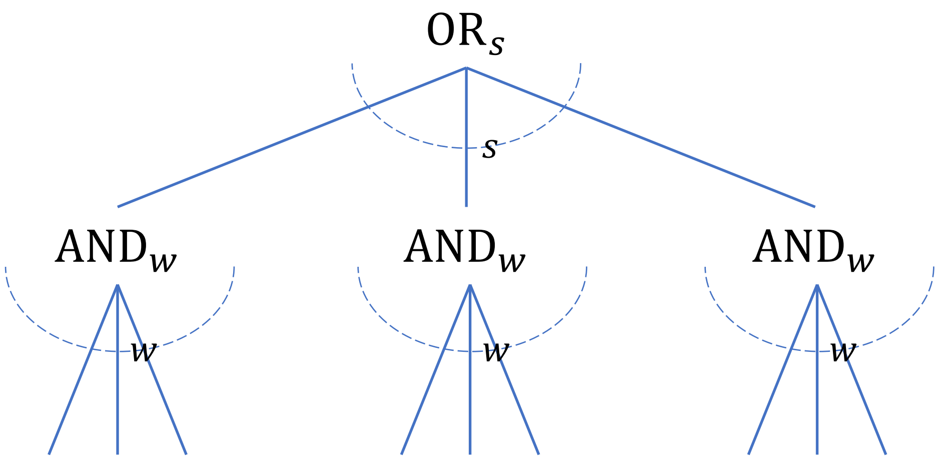

Desirable properties

Monotone: $f(x) \leq f(y)$ whenever $x_i \leq y_i$ for all i

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	Maj_n (n odd)	χ_i	AND_n / OR_n	$\mathrm{Tribes}_{w,s}$
Monotone	✔	✔	✔	✔
Odd	✔	✔	✘	✘
Unanimous	✔	✔	✔	✔
Symmetric	✔	✘	✔	✘

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Unanimous	✓	✓	✓	✓
Symmetric	✓	✗	✓	✗

Exercise

May's theorem: Maj_n is the only monotone, odd, symmetric voting rule.

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Unanimous	✔	✔	✔	✔
Symmetric	✔	✘	✔	✘

$\mathrm{Tribes}_{w,s}$ is **transitive-symmetric**:

- for all $i, i' \in [n]$, there is a permutation $\pi \in S_n$ with $\pi(i) = i'$ such that:
  - for all $x \in \{-1,1\}^n$, $f(x^\pi) = f(x)$

Influence of a voter

$x^{\oplus i} = (x_1, \dots, x_{i-1}, -x_i, x_{i+1}, \dots, x_n)$

$i \in [n]$ is **pivotal** for $f : \{-1,1\}^n \to \{-1,1\}$ on input x if $f(x) \neq f\left(x^{\oplus i}\right)$

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The **influence** of a coordinate i on f is $\underbrace{\mathrm{Inf}_i[f] = \Pr_{\boldsymbol{x} \sim \{-1,1\}^n}[i \text{ is pivotal for } f] = \mathbb{E}_{\boldsymbol{x} \sim \{-1,1\}^n}\left[f(x) \neq f\left(x^{\oplus i}\right)\right]}$

Impartial cultural assumption: $\boldsymbol{x} \sim \{-1,1\}^n$

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Geometric interpretation:
$\mathrm{Inf}_i[f]$ is the fraction of the dimension-i edges that cross the cut represented by f in the n-dimensional hypercube.

Influence of a voter

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$$\mathrm{Inf}_i[\chi_i] = \begin{cases} 1 & i = 1 \\ 0 & i \neq 1 \end{cases}$$

$$\mathrm{Inf}_i[\mathrm{AND}_n] = \mathrm{Inf}_i[\mathrm{OR}_n] = 2^{1-n}$$

$$\mathrm{Inf}_i[\mathrm{Maj}_n] \in \sqrt{\frac{2}{\pi n}} + O(n^{-1.5})$$

Exercises

$\mathrm{Inf}_i[f] \leq 1/\sqrt{n}$ if f is **transitive-symmetric** and **monotone**. ← **Next**

Influence of a voter

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$x^{i \mapsto b} = (x_1, \dots, x_{i-1}, b, x_{i+1}, \dots, x_n)$

Derivative operator:

$$\mathrm{D}_i f(x) = \frac{f\left(x^{i \mapsto 1}\right) - f\left(x^{i \mapsto -1}\right)}{2}$$

$$\mathrm{Inf}_i[f] = \mathbb{E}_{\boldsymbol{x} \sim \{-1,1\}^n}\left[\left(\mathrm{D}_i f(x)\right)^2\right] = \|\mathrm{D}_i f(x)\|_2^2$$

The generalization of influence to any real-valued function $f : \{-1,1\}^n \to \mathbb{R}$

Influence of a voter

Claim: $\mathrm{Inf}_i[f] = \sum_{S \ni i} \hat{f}(S)^2$

Parseval's theorem

$$\mathrm{D}_i \chi_S = \begin{cases} \chi_{S \setminus \{i\}} & i \in S \\ 0 & i \notin S \end{cases} \longrightarrow \mathrm{D}_i f(x) = \sum_{S \ni i} \hat{f}(S)\, \chi_{S \setminus \{i\}}$$

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$f : \{-1,1\}^n \to \{-1,1\}$ is **monotone**

$$\mathrm{Inf}_i[f] = \mathop{\mathbb{E}}_{\boldsymbol{x} \sim \{-1,1\}^n}[\mathrm{D}_i f(x)] = \widehat{\mathrm{D}_i f}(\emptyset) = \hat{f}(\{i\})$$

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$\color{red}{\mathrm{Inf}_i[f] \le 1/\sqrt{n}}$

$$1 = \sum_{S\subseteq[n]} \hat{f}(S)^2 \ge \sum_{i\in[n]} \hat{f}(\{i\})^2 = n\cdot(\mathrm{Inf}_i[f])^2$$

f is **transitive-symmetric**

$$\boxed{x^{i\mapsto b} = (x_1, \dots, x_{i-1}, \color{red}{b}, x_{i+1}, \dots, x_n)}$$

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$$\begin{aligned} &\mathrm{Inf}_i[f] \\ &= \mathop{\mathbb{E}}_{x\sim\{-1,1\}^n}[\mathrm{D}_i f(x)] \\ &= \widehat{\mathrm{D}_i f}(\emptyset) \\ &= \hat{f}(\{i\}) \end{aligned}$$

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The generalization of influence to any real-valued function $f : \{-1,1\}^n \to \mathbb{R}$

Total influence

Total influence: $\mathrm{I}[f] = \sum_{i \in [n]} \mathrm{Inf}_i[f]$

Geometric interpretation:
$\mathrm{I}[f]$ is the fraction of the edges that cross the cut represented by f in the n-dimensional hypercube.

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Suppose $f : \{-1,1\}^n \to \{-1,1\}$ is **monotone**.

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Observation: Under the Impartial cultural assumption, the expected number of votes that agree with the outcome of the election is

$$\frac{n}{2} + \frac{1}{2} \cdot \sum_{i\in[n]} \hat{f}(\{i\})$$

$$\sum_{i\in[n]} \hat{f}(\{i\}) = \sum_{i\in[n]} \mathop{\mathbb{E}}_{\boldsymbol{x}\sim\{-1,1\}^n}[f(\boldsymbol{x})\boldsymbol{x}_i] = \mathop{\mathbb{E}}_{\boldsymbol{x}\sim\{-1,1\}^n}[f(\boldsymbol{x})\underbrace{(\boldsymbol{x}_1 + \cdots + \boldsymbol{x}_n)}_{\#\,(\text{votes for winner}) \,-\, \#\,(\text{votes for loser})}]$$

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Corollaries:

- Maj_n is the unique maximizer of $\sum_{i\in[n]} \hat{f}(\{i\})$ over all functions $f : \{-1,1\}^n \to \{-1,1\}$.
- $\mathrm{I}[f] \le \mathrm{I}[\mathrm{Maj}_n]$ for any **monotone** $f : \{-1,1\}^n \to \{-1,1\}$.

Total influence

Total influence: $\mathrm{I}[f] = \sum_{i \in [n]} \mathrm{Inf}_i[f]$

$$\mathrm{Inf}_i[f] = \sum_{S \ni i} \hat{f}(S)^2 \longrightarrow \mathrm{I}[f] = \sum_{S \subseteq [n]} |S| \hat{f}(S)^2$$

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Total influence: $\mathrm{I}[f] = \sum_{i\in[n]} \mathrm{Inf}_i[f]$

$$\mathrm{Inf}_i[f] = \sum_{S\ni i} \hat{f}(S)^2 \longrightarrow \underbrace{\mathrm{I}[f] = \sum_{S\subseteq[n]} |S|\hat{f}(S)^2 \geq \sum_{S\subseteq[n]\setminus\emptyset} \hat{f}(S)^2 = \mathrm{Var}[f]}_{\textbf{Poincaré inequality}}$$

Exercise: The conductance of n-vertex hypercube is $\Omega(1/\log n)$.

Outlook

- For 2-candidate voting, it is quite clear that **majority** is the right choice.
  - May's theorem.
  - $\mathrm{I}[f] \leq \mathrm{I}[\mathrm{Maj}_n]$ for any monotone $f : \{-1,1\}^n \to \{-1,1\}$.
- Things become more complicated when we have more than 2 candidates.
- **Next:** Arrow's impossibility theorem.

When there are more than 2 candidates, no voting system can convert individual preferences into a collective decision while satisfying a set of seemingly reasonable fairness conditions simultaneously.

References

- Sections 2.1–2.3 of https://arxiv.org/abs/2105.10386